

StockWISE Handoff — Production Hosting

This document is for engineers taking over StockWISE hosting. It covers deploying the **frontend on Vercel** and the **backend plus PostgreSQL on a fresh Contabo Ubuntu VPS**.

Placeholder URLs used below

Role	Placeholder
Frontend (Vercel)	<code>https://your-frontend.vercel.app</code>
Backend (Contabo)	<code>https://api.yourdomain.com</code>
Database (Contabo, localhost only)	<code>postgresql://stockwise:<password>@127.0.0.1:5432/stockwise</code>

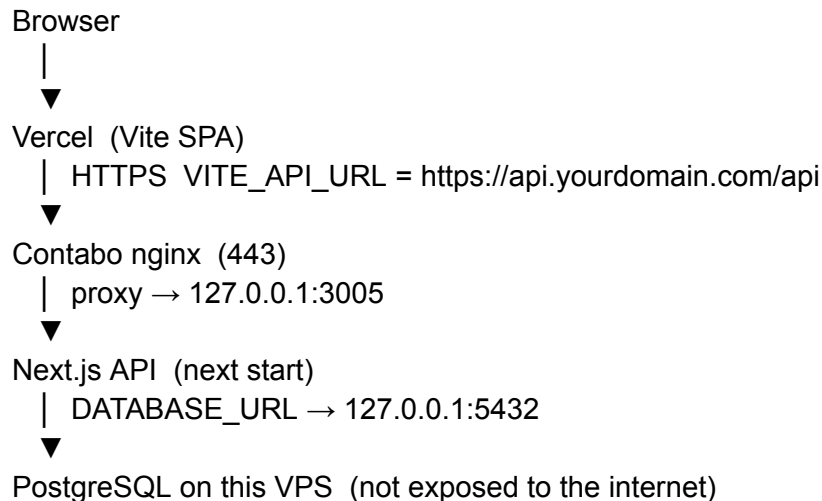
Replace these with your real URLs and password. Do not copy secrets into this file or commit `.env` / `.env.local`.

1. What you are taking over

StockWISE is a staff-facing stock and order app (GHS / Ghana trade). Three services make production work — all data stays on the VPS:

Service	What it is	Where it lives
Frontend	React 18 + Vite SPA (React Router, Tailwind)	Vercel, project root = <code>frontend</code>

Backend	Next.js 14 Pages Router API (<code>npm start = next start</code>)	Contabo VPS behind nginx
Database	PostgreSQL (tables only)	Same Contabo VPS, listen on <code>127.0.0.1:5432</code> only



Rules

- The frontend never talks to the database. All reads/writes go through the API.
- Install PostgreSQL **on Contabo**. Do not use a hosted database service.
- Do **not** open Postgres (`5432`) or the Node port (`3005`) on the public firewall. nginx (`80/443`) and SSH (`22`) only.
- No Docker, Express rewrite, or extra native packages (`sharp`, `puppeteer`) are required.
- Login is UI-only: the frontend stores `stockwise-user` in `localStorage`. CRUD API routes are **not** JWT-protected. Treat the public API as open unless you add hardening later.

Older READMEs mention backend ports `3000` or `3001`. For production, use `PORT=3005`.

2. Access checklist

Get these from the previous owner before you start. Do not store passwords in the repo.

Access

Why you need it

GitHub repo	Clone on Contabo; connect Vercel
Vercel account	Frontend project
Contabo VPS	SSH as a sudo user, public IPv4
Domain DNS	A record for <code>api.yourdomain.com</code> → VPS IP
Postgres password	You create this on the VPS (see §5.4).

Production uses `DATABASE_URL` pointing at local PostgreSQL. Copy frontend names from `frontend/.env.example`. For the backend, set the variables listed in §3 — do not copy hosted-database URLs from older notes.

3. Environment variables

Never commit real values. Vite inlines `VITE_*` at **build** time — changing a frontend env var requires a Vercel **redploy**.

Frontend (Vercel → Settings → Environment Variables)

Variable	Required	Production value
<code>VITE_API_URL</code>	Yes	<code>https://api.yourdomain.com/api</code> (must include <code>/api</code> , no trailing slash after <code>api</code>)
<code>VITE_DEBUG</code>	No	<code>1</code> only while debugging a production build

VITE_ENQUIRY_PHONE	No	Click-to-call number (Ghana local 0XXXXXXXX)
--------------------	----	--

VITE_WHATSAPP_PHONE	No	Digits + country code, no + (e.g. 233...)
---------------------	----	---

VITE_SUPPORT_PHONE	No	Legacy alias for enquiry phone
--------------------	----	--------------------------------

Backend (Contabo file: **backend/.env.local**)

Variable	Required	Production value
PORT	Yes	3005 (nginx proxies here; do not open 3005 on the public firewall)
DATABASE_URL	Yes	postgresql://stockwise:<password>@127.0.0.1:5432/stockwise — local Postgres only
ALLOWED_ORIGINS	Yes	Exact frontend origin, no trailing slash. Put the Vercel URL first if you list more than one. Example: https://your-frontend.vercel.app
LOG_LEVEL	No	info (or debug while diagnosing)
RATE_LIMIT_MAX / RATE_LIMIT_WINDOW_MS	No	Defaults: 2000 requests / 15 minutes per IP

<code>STOCKWISE_RUN_MIGRATIONS</code>	No	<code>true</code> only if you want <code>npm run build</code> to migrate (see §5.6)
<code>STOCKWISE_SKIP_MIGRATIONS</code>	No	<code>true</code> to force-skip the build migration hook
<code>JWT_SECRET</code> / <code>JWT_EXPIRES_IN</code>	Unused	Placeholders only — not applied today

CORS note: Runtime CORS in `backend/lib/middleware.js` checks the full `ALLOWED_ORIGINS` list. The static header in `backend/next.config.js` uses only the **first** entry. Put `https://your-frontend.vercel.app` first.

4. Order of operations

The API URL and CORS origin depend on each other. Do this in order:

1. Point DNS at the VPS.
2. Install **PostgreSQL** on Contabo, create the database, apply the schema (§5.3–5.6).
3. Deploy the **backend** on Contabo (build, PM2, nginx, TLS).
4. Confirm `curl https://api.yourdomain.com/api/health` returns healthy.
5. Deploy the **frontend** on Vercel with `VITE_API_URL=https://api.yourdomain.com/api` (steps in §6).
6. Set backend `ALLOWED_ORIGINS` to the exact Vercel URL (no trailing slash), then restart the API (`pm2 restart stockwise-api`).
7. Smoke-test login and one list page (§7).

If you deploy the frontend first, you can still set `VITE_API_URL` after the API exists — but you **must redeploy** the frontend for Vite to pick it up.

5. Backend and PostgreSQL on Contabo (fresh Ubuntu)

Assumes a new Ubuntu 22.04/24.04 VPS, a sudo user, and a domain you control. Commands below use `/opt/stockwise` as the app directory.

5.1 DNS

Create an A record:

api.yourdomain.com → <Contabo VPS IPv4>

Wait until `dig api.yourdomain.com` (or your DNS panel) shows the VPS IP.

5.2 Packages

```
sudo apt update
sudo apt install -y git nginx certbot python3-certbot-nginx ufw curl postgresql
postgresql-contrib
```

Install Node.js 20 LTS (Next.js 14 needs Node 18.17+; 20 LTS is fine):

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
node -v # v20.x
npm -v
```

Install PM2 globally:

```
sudo npm install -g pm2
```

5.3 Firewall

Do **not** expose port `3005` or Postgres `5432` to the internet.

```
sudo ufw allow OpenSSH
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
sudo ufw enable
sudo ufw status
```

Confirm `5432` is not listed as allowed.

5.4 Install and create PostgreSQL

```
sudo systemctl enable --now postgresql
sudo systemctl status postgresql
```

Create a dedicated role and database. Pick a strong password and keep it only in `.env.local`.

```
sudo -u postgres psql
```

In the `psql` prompt:

```
CREATE USER stockwise WITH PASSWORD 'choose-a-strong-password';
CREATE DATABASE stockwise OWNER stockwise;
GRANT ALL PRIVILEGES ON DATABASE stockwise TO stockwise;
```

\q

Confirm Postgres is listening on localhost only:

```
sudo ss -ltnp | grep 5432
# expect 127.0.0.1:5432 (or ::1), not 0.0.0.0:5432
```

If it is bound to all interfaces, edit `postgresql.conf` (`listen_addresses = 'localhost'`), reload Postgres, and leave `pg_hba.conf` allowing `scram-sha-256` (or `md5`) for `127.0.0.1/32` only.

Test the URL you will put in `.env.local`:

```
psql "postgresql://stockwise:choose-a-strong-password@127.0.0.1:5432/stockwise" -c
'SELECT version();'
```

5.5 Clone the app and create `backend/.env.local`

```
sudo mkdir -p /opt/stockwise
sudo chown "$USER":"$USER" /opt/stockwise
git clone <repo-url> /opt/stockwise
cd /opt/stockwise/backend
npm ci
nano /opt/stockwise/backend/.env.local
PORT=3005
DATABASE_URL=postgresql://stockwise:choose-a-strong-password@127.0.0.1:5432/stock
wise
ALLOWED_ORIGINS=https://your-frontend.vercel.app
LOG_LEVEL=info
```

`chmod 600 /opt/stockwise/backend/.env.local` so only your user can read it.

If you do not know the Vercel URL yet, set `ALLOWED_ORIGINS` to a placeholder and update it after §6, then restart PM2.

5.6 Database schema (do this once)

Apply `backend/database/migrate-all.sql` to the **local** database. Prefer `psql` on the VPS.

```
psql "$DATABASE_URL" -f /opt/stockwise/backend/database/migrate-all.sql
```

Or, if `DATABASE_URL` is not exported in the shell:

```
set -a
source /opt/stockwise/backend/.env.local
set +a
psql "$DATABASE_URL" -f /opt/stockwise/backend/database/migrate-all.sql
```

You can also run `cd /opt/stockwise/backend && npm run migrate:all`. If that fails with an SSL error against localhost, use the `psql` command above — same SQL file.

Confirm tables:

```
psql "$DATABASE_URL" -c "\dt"
```

You should see `suppliers`, `products`, `customers`, `orders`, `order_items`, `users`.

`npm run build` runs `migrate:deploy`, which only applies SQL when `VERCEL=1` or `STOCKWISE_RUN_MIGRATIONS=true`. On Contabo, apply schema with `psql` (or `migrate:all`) yourself, then skip the hook on later builds.

Skip `npm run seed` in production unless you are asked to load demo data. The seed script does not load dotenv — export `DATABASE_URL` in the shell if you run it.

5.7 Build

After the schema exists:

```
cd /opt/stockwise/backend
STOCKWISE_SKIP_MIGRATIONS=true npm run build
```

To migrate during build instead (needs `DATABASE_URL`; may fail on localhost SSL — use `psql` if so):

```
STOCKWISE_RUN_MIGRATIONS=true npm run build
```

5.8 Start with PM2

```
cd /opt/stockwise/backend
# next start reads PORT from .env.local
pm2 start npm --name stockwise-api -- start
pm2 save
pm2 startup
# run the command `pm2 startup` prints (sudo env PATH=...)
```

Local check (from the VPS):

```
curl http://127.0.0.1:3005/api/health
```

Expected JSON includes `"status": "healthy"`. Health does not prove the database is wired; `/api/suppliers` does.

5.9 nginx reverse proxy

```
sudo nano /etc/nginx/sites-available/stockwise-api
server {
    listen 80;
```

```
server_name api.yourdomain.com;

location / {
    proxy_pass http://127.0.0.1:3005;
    proxy_http_version 1.1;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
}
}
```

`X-Forwarded-For` and `X-Forwarded-Proto` are required. The API rate limiter keys on client IP from that header.

```
sudo ln -s /etc/nginx/sites-available/stockwise-api /etc/nginx/sites-enabled/
sudo nginx -t
sudo systemctl reload nginx
```

5.10 HTTPS

DNS for `api.yourdomain.com` must already resolve to this VPS.

```
sudo certbot --nginx -d api.yourdomain.com
```

Verify:

```
curl https://api.yourdomain.com/api/health
```

6. Frontend on Vercel

1. In vercel.com, **Add New Project** and import the StockWISE GitHub repo.
2. Set **Root Directory** to `frontend` (monorepo — do not deploy the repo root).
3. Confirm:
 - Framework Preset: **Vite**
 - Build Command: `npm run build`
 - Output Directory: `dist`
 - Install Command: `npm install` (default)
4. **Environment Variables** (Production, and Preview if you use preview URLs):
 - VITE_API_URL=https://api.yourdomain.com/api
 - Optional: `VITE_ENQUIRY_PHONE`, `VITE_WHATSAPP_PHONE`.
5. Deploy.

SPA routing is already handled by frontend/vercel.json (all paths rewrite to `index.html`). A refresh on `/products` should load the app, not a Vercel 404.

After the first deploy, copy the production URL (e.g. <https://your-frontend.vercel.app>). Put that exact origin first in backend `ALLOWED_ORIGINS`, then:

```
# on the VPS
nano /opt/stockwise/backend/.env.local
# edit ALLOWED_ORIGINS, save
pm2 restart stockwise-api
```

If you change `VITE_API_URL` later, trigger a **new Vercel deployment**. Editing the env var alone is not enough.

7. Smoke test

Run these after both sides are up.

```
curl https://api.yourdomain.com/api/health
curl https://api.yourdomain.com/api/suppliers
psql "$DATABASE_URL" -c "SELECT COUNT(*) FROM suppliers;"
```

In the browser:

1. Open <https://your-frontend.vercel.app>.
2. Sign up or log in. After success, DevTools → Application → Local Storage should contain `stockwise-user`.
3. Open Products or Suppliers and confirm a list loads (or an empty state from the API, not a network/CORS error).
4. Hard-refresh `/products` — the page should still load (SPA rewrite).

Common failures

Symptom	Likely cause	Fix
<code>Failed to fetch</code> / CORS error	Frontend origin not in <code>ALLOWED_ORIGINS</code> , or trailing slash mismatch	Set exact origin, restart PM2
“API URL not configured” / empty <code>API_BASE</code>	<code>VITE_API_URL</code> missing at build time	Set it on Vercel and redeploy

Login / list routes error after a healthy <code>/api/health</code>	<code>DATABASE_URL</code> wrong, Postgres down, or schema not applied	Recheck §5.4–5.6, <code>pm2 logs stockwise-api</code>
<code>npm run migrate:all</code> SSL error	Node migrator expecting SSL on localhost	Apply SQL with <code>psql "\$DATABASE_URL" -f backend/database/migrate-all.sql</code>
<code>psql</code> password / connection refused	Wrong <code>DATABASE_URL</code> , or Postgres not listening on <code>127.0.0.1</code>	Recheck §5.4
Login POST returns 405	Process not serving API routes, or missing env	Check <code>.env.local</code> , <code>pm2 logs stockwise-api</code>
Health works, browser still fails	<code>VITE_API_URL</code> missing <code>/api</code> , or still pointing at localhost / old Vercel backend	Fix env, redeploy frontend
404 on <code>/products</code> refresh	Root Directory is not <code>frontend</code> , or <code>vercel.json</code> not deployed	Confirm Vercel settings

More CORS / 405 background: backend/VERCEL_DEPLOY.md (ignore steps that deploy the backend to Vercel).

8. Day-2 operations

Redeploy frontend

Push to the branch Vercel tracks. Vercel rebuilds **frontend**. If you only changed env vars, use **Redeploy** in the Vercel dashboard.

Redeploy backend

```
cd /opt/stockwise
git pull
cd backend
npm ci
STOCKWISE_SKIP_MIGRATIONS=true npm run build
pm2 restart stockwise-api
```

When the schema changes, apply SQL to local Postgres with **psql**:

```
set -a && source /opt/stockwise/backend/.env.local && set +a
psql "$DATABASE_URL" -f /opt/stockwise/backend/database/migrate-all.sql
```

Database backups

```
sudo mkdir -p /opt/backups
sudo chown "$USER":"$USER" /opt/backups
set -a && source /opt/stockwise/backend/.env.local && set +a
pg_dump "$DATABASE_URL" > /opt/backups/stockwise-$(date +%F).sql
```

Restore (destroys current data — only when you intend to):

```
psql "$DATABASE_URL" < /opt/backups/stockwise-YYYY-MM-DD.sql
```

Logs

```
pm2 logs stockwise-api
pm2 status
sudo journalctl -u nginx -f
sudo journalctl -u postgresql -f
sudo tail -f /var/log/nginx/error.log
```

Rotate secrets

1. Change the Postgres password in **psql**, then update **DATABASE_URL** in **/opt/stockwise/backend/.env.local** (never commit it).
2. **pm2 restart stockwise-api**.
3. For frontend secrets (**VITE_***), change them in Vercel and **redeploy**.

Process after reboot

pm2 startup + **pm2 save** should bring **stockwise-api** back. **postgresql** is a systemd service. Confirm with **pm2 status**, **sudo systemctl status postgresql**, and **curl http://127.0.0.1:3005/api/health**.

9. Production caveats (not part of this hosting task)

These are current product behaviors. Do not treat them as deploy blockers.

- API routes do not validate a session token. Anyone who can reach <https://api.yourdomain.com/api> can call CRUD endpoints.
- All operators share one data set (no multi-tenant isolation).
- `migrate-all.sql` enables RLS with permissive allow-all policies. Connections as the table owner (typical for local `psql` / `DATABASE_URL`) bypass RLS. See `APP_DOCUMENTATION.md` §12.